Transaction Ledger API

Assumptions

1. Account balance starts at zero

- The initial account balance is always **0**.
- The client cannot provide or modify the initial balance when creating an account.

2. Reference ID is generated by the backend

- The `reference_id` is automatically generated by the backend for every transaction.
- The client does not need to provide a `reference_id`.
- Each generated reference ID is unique.

3. Transaction description is optional

- A transaction can be created without a description.
- If no description is provided, an empty description is stored.

4. Transaction amounts must be positive

- Transaction amounts must be greater than **0**.
- Zero and negative amounts are rejected.

5. Debit transactions require sufficient balance

- A debit is rejected when the account balance is less than the requested amount.
- A failed debit does not modify the account balance or create a transaction.

6. Transactions are immutable

- Transactions cannot be edited or deleted through the API after creation.

7. Balance and transaction creation are atomic

- Creating a transaction and updating the account balance are performed within the same MongoDB transaction.
- If any operation fails, the complete operation is rolled back.

8. Transfers are atomic

- A transfer consists of a debit from the source account and a credit to the destination account.
- Both account balance updates and both transaction records are handled within the same MongoDB transaction.
- If any part of the transfer fails, the entire transfer is rolled back.

9. Source and destination accounts must be different

- An account cannot transfer money to itself.

10. Transaction history is latest-first

- Transaction history is sorted by `created_at` in descending order.

11. Pagination defaults

- Transaction history defaults to page **1** with a page size of **10**.
- The maximum page size is limited to **100**.

12. Authentication is not implemented

- The assessment requirements do not specify user authentication or authorization.
- Therefore, authentication and passwords are not included in the implementation.

13. MongoDB Atlas is used as the database

- The application uses MongoDB Atlas for persistence.
- MongoDB transactions are used to maintain consistency between balances and transactions.

14. No currency conversion

- The application treats all transaction amounts as numeric monetary values.
- Currency conversion and multi-currency support are outside the scope of this assessment.

15. No transaction editing or deletion

- Once a transaction is successfully created, it remains part of the transaction history permanently.